Course outline

Syllabus:

Course Code: 1929 **Name of the Programme: Skill Enhancement** **Name of the Course: Structured Innovation with Design Thinking** **Semester: I**		
Course credit	**No. of hours per week**	**Total no. of Teaching hours**
2	1:0:2	45
Course Objectives:		
● Apply design thinking for creative problem-solving		
● Synthesize concepts for sustainable prototype development		
● Enhance existing products focusing on market aesthetics		
● Innovate within frameworks to create new prototypes and communicate		

Module - 1 Design thinking for creative problem-solving	**No. of hours**
	10
Introduction to Design Thinking: Overview of design thinking principles and stages. Empathy in Design: Techniques for understanding and empathizing with users, including user interviews and persona creation. Storytelling as a Tool: Using storytelling to convey user needs and problem statements. Case Studies: Analyzing successful design thinking projects that started with empathy. Workshops: Hands-on activities to practice empathic engagement with hypothetical user scenarios.	

Module - 2 Concepts for sustainable prototype development	**No. of hours**
	10
Basics of Recycling and Upcycling: Definitions, differences, and environmental impact. Triple Constraints in Product Design: Exploring how time, scope, and budget affect product development. Prototype Development: Step-by-step guidance on creating prototypes using upcycled materials. Sustainability in Design: Discussing the importance of sustainable practices in modern design. Practical Exercises: Designing and building simple products using recycled materials within defined constraints.	

Module - 3 Existing products focusing on market aesthetics	**No. of hours**
	10
Understanding Product Aesthetics: How aesthetics influence consumer behavior and product design. Detailed Product Analysis: Dismantling and evaluating existing products to understand their design and functionality. Enhancement Techniques: Methods for improving product design, focusing on aesthetics and functionality. Consumer Feedback: Gathering and utilizing consumer feedback to guide redesign efforts. Reconstruction Projects: Students reconstruct an existing product with improved aesthetics and functionality.	

Module - 4 Frameworks to create new prototypes	**No. of hours**
	10
Innovation vs. Invention: Exploring the difference and focusing on innovation within existing frameworks. Analyzing Market Trends: How to use market research to identify opportunities for innovation. Creative Adaptations: Techniques for creative thinking and adaptation of existing products. Prototyping with Purpose: Building prototypes that address specific user needs or market gaps. Case Studies and Discussions: Reviewing examples of successful innovations in existing product lines.	

Module - 5 Communicate project processes and outcomes	**No. of hours**
	5
The Art of Storytelling: Techniques for crafting compelling narratives around product development. Presentation Skills: Tips and strategies for effective public speaking and presentation. Visual and Digital Tools: Using visual aids, digital presentations, and storytelling software to enhance communication. Engaging Stakeholders: How to pitch and explain projects to different audiences (investors, customers, peers). Project Documentation and Feedback: Methods for documenting the development process and incorporating feedback for improvement.	

Lab Plan	**[30 Hours]**
1	Day 1: Empathy Activity & Paper Prototyping
2	Day 2: Define & Ideate Activity
3	Day 2: Upcycling
4	Day 3: Product study
5	Day 3: Product variation & Testing
6	Day 4: Product development: Game Design for a purpose

Course outcomes
CO1 Apply design thinking for creative problem-solving
CO2 Synthesize concepts for sustainable prototype development
CO3 Enhance existing products focusing on market aesthetics
CO4 Innovate within frameworks to create new prototypes and communicate

Text books	
1.	Jake Knapp, John Zeratsky, and Braden Kowitz, "Sprint: How to Solve Big Problems and Test New Ideas in Just Five Days" Simon & Schuster (2016), ISBN: 978-1501121746, 1st ed.
2.	Idris Mootee "Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School" Wiley (2013), ISBN: 978-1118620120, 1st ed.

Reference books	
1.	Jeanne Liedtka and Tim Ogilvie "Designing for Growth: A Design Thinking Toolkit for Managers" Columbia University Press (2011) ISBN: 978-0231158381, 1st ed.
2.	Pavan Soni, Design Your Thinking, Penguin Random House India Private Limited (2020), 978-0670094097, 1st ed.

Lesson plan

Name of the Programme:	Skill Enhancement	
Title of the Course:	Structured Innovation with Design Thinking	
Course code	CS1929	
Semester:	I	
Names of the Course Instructor(s)	Prof. Chidhananda R S Prof. Chethan Kumar G Prof. N Suresh	
Course credit:	No. of hours per week (1+0+2)	Total no. of Teaching hours
02	03 Hours	45 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	1	Introduction to Design Thinking	Lecture + Case Videos	Knapp et al. (Ch. 1), Mootee (Ch. 1)
2.		Empathy Mapping & User Interviews	Empathy activities + Role play	Liedtka & Ogilvie (Ch. 1)
3.		Persona Creation & Storytelling	Group work + Scenario building	Soni (Ch. 2)
4.		Design Thinking with Artificial Intelligence (AI)	Exploratory discussion + AI-based design tools demo	Research articles, Online AI toolkits
5.		Activity: Paper Prototyping	Hands-on session	Provided templates
6.	2	Introduction to Sustainability in Design	Video + Comparative study	Mootee (Ch. 3)
7.		Basics of Recycling and Upcycling	Demonstration + Product tear-down	Mootee (Ch. 4), Online sources
8.		Triple Constraints in Design	Group design challenge	Knapp (Ch. 3)
9.		Activity: Upcycled Product Prototype	Material-based prototyping	Local case examples
10.	3	Understanding Product Aesthetics	Slide presentation + Object analysis	Industrial product examples

		Dismantling and Evaluating Products	Hands-on session	Consumer product references
12.		Gathering Consumer Feedback	Peer reviews + Survey analysis	Online tools, Soni (Ch. 5)
13.		Reconstructing with Market Aesthetics	Redesign activity	Workshop materials
14.	**4**	Innovation vs. Invention	Debate + Market trend insights	Knapp (Ch. 4), Trendwatching.com
15.		Market Research for Innovation	Data-based ideation	Templates provided
16.		Creative Thinking for Adaptations	Brainstorming + Creative matrix	Soni (Ch. 3)
17.		Activity: Purposeful Prototyping	Problem-driven prototype design	Workshop brief
18.	**5**	Storytelling & Pitch Craft	Design narrative development	Liedtka (Ch. 4)
19.		Visual and Digital Tools for Communication	Canva, Google Slides, Whiteboarding	Online tutorials
20.		Final Project Presentation	Team presentations + Peer Q&A	All course material
21.		Stakeholder Feedback and Iteration Planning	Peer critique + Feedback synthesis and action mapping	Soni (Ch. 6), Project logs

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Empathy Mapping and Prototyping for Paper Prototyping Design - Upcycling for 10x Value Creation & Sustainability - User Persona Interviews and Ideation	10 marks out of 70 5 marks out of 70 5 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Prototype design using AI - Existing Product Study & Variation - Creating a New Game Product	5 marks out of 70 5 marks out of 70 15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Jury
Weightage of exam	30%

Course Outcomes- Progamme Outcomes Matrix

Program Outcome Course Outcome		PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO7	PO 8	PO9	PO 10	PO 11	PO 12
CO1	CO1 Apply design thinking for creative problem-solving	3	2	3		3	1						
CO2	CO2 Synthesize concepts for sustainable prototype development	3	2	3		3							
CO3	CO3 Enhance existing products focusing on market aesthetics	3	2	3	2	2	1						
CO4	CO4 Innovate within frameworks to create new prototypes and communicate	3	2	3	2	2	1	1	1				

Rubrics for evaluation of CIE (separate for each CIE component)

<table>
<tr><th colspan="2">Total Marks</th><th>20</th></tr>
<tr><th colspan="2">CIE-1 Components</th><th>Rubrics for Assessment</th></tr>
<tr><td rowspan="3">CIE-1</td><td>Empathy Mapping and Prototyping for Greeting Card Design - Graded Component 1</td><td>10 Marks, Students create empathy maps and paper prototypes based on observed user pain points. Table-1</td></tr>
<tr><td>Upcycling for 10x Value Creation & Sustainability - Graded Component-2</td><td>5 Marks, Design an upcycled product using discarded materials with a focus on value innovation and sustainability. Table-2</td></tr>
<tr><td>Prototype design using AI - Graded Component 3</td><td>5 Marks, Conduct short interviews, define user personas, and ideate problem-solution pairs. Table-3</td></tr>
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Table-1

Component 1: Empathy Mapping & Prototyping

Criteria	Excellent (9–10 M)	Very Good (7–8 M)	Good (5–6 M)	Needs Improvement (0–4 M)
Empathy Mapping	Deep insights into user needs	Clear, relevant, minor gaps	Basic mapping	Vague or disconnected
Paper Prototype	Functional and user-aligned	Logical but less refined	Basic sketch	Unclear or incomplete

Table-2

Component 2: Upcycling Product Design

Criteria	Excellent (5 M)	Very Good (4 M)	Good (3 M)	Needs Improvement (0–2 M)
Innovation	Highly creative reuse	Creative and functional	Functional but common	Minimal reuse or effort
Sustainability Relevance	Clearly linked to sustainability	Moderate impact	Weak linkage	No relevance shown

Table-3

Component 3: User Persona & Ideation

Criteria	Excellent (5 M)	Very Good (4 M)	Good (3 M)	Needs Improvement (0–2 M)
Persona Accuracy	Specific, realistic, data-backed	Clear persona with good details	Generic persona	Unclear or fabricated persona
Ideation Quality	Novel, relevant ideas with user alignment	Practical with some creativity	Basic ideas	Random or disconnected ideas

Total Marks		**25**
CIE-2 Components		**Rubrics for Assessment**
CIE-2	Theory	The test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the team of course faculty.

Total Marks		**25**
CIE-3 Components		**Rubrics for Assessment**
1	**AI-based Prototype Design**	5 Marks, Develop a product or feature prototype using AI tools (e.g., ChatGPT, Midjourney, Figma AI, etc.) ***Table-4***
2	**Existing Product Study & Variation**	5 Marks, Select a real-world product, analyze pain points, and propose a variation or improvement. ***Table-5***
3	**New Game/Product Creation**	15 Marks, Ideate, design, and document a new game or product concept, with focus on innovation and user value. ***Table-6***

Table-4

Component 1: AI-based Prototype Design

Criteria	Excellent (5 M)	Very Good (4 M)	Good (3 M)	Needs Improvement (0–2 M)
Tool Usage	Effective use of AI tools for design	Minor issues in tool use	Basic use of AI, less refined	Ineffective or unclear use
Output Quality	Prototype is novel and well-presented	Good structure, lacks polish	Functional but basic	Incomplete or unclear output

Table-5

Component 2: Existing Product Study & Variation

Criteria	Excellent (5 M)	Very Good (4 M)	Good (3 M)	Needs Improvement (0–2 M)
Analysis Depth	Deep evaluation with insights	Clear analysis with minor gaps	Basic review	Superficial or copied
Variation Relevance	Clear improvement addressing gaps	Logical but not innovative	Some relevance	No improvement proposed

Table-6

Component 3: New Game/Product Creation

Criteria	Excellent (13–15 M)	Very Good (10–12 M)	Good (7–9 M)	Needs Improvement (0–6 M)
Originality & Creativity	Highly innovative concept	Fresh idea with user alignment	Functional but generic	Unoriginal or unclear
Documentation	Well-structured, visuals, flowcharts	Mostly complete	Basic structure	Incomplete
User-Centric Design	Strong focus on user needs	Moderate alignment	Basic consideration	Lacks user perspective

Signature of the Course Faculty